Supplementary Information for:

No Selves, No Consciousness

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This document accompanies the main paper *No Selves, No Consciousness* (AAAI 2026 Spring Symposium on Machine Consciousness). It contains background definitions, full proofs of all theorems stated in the main paper, additional result tables, and detailed experimental methods.

A Additional results

A.1 Experiment 1 sensitivity sweep

Table 1: Experiment 1 sensitivity to uneven priors. Each row changes the Beta(a, b) settings used to draw random chances for unseen outputs.

Beta a	Beta b	Mean regret w max	Mean regret simp max
0.5	2.5	0.200	4.370
0.7	2.0	0.152	7.301
1.0	1.0	0.696	19.132
2.0	0.7	0.294	38.677

A.2 Experiment 3 trust with promise types

Table 2: Experiment 3 trust rates at horizon $H = 1$ with honest and liar promise types. Mean is across worlds and low and high give a 95% range.

Mode	Mean	95% CI low	95% CI high
silent	0.000	0.000	0.000
promise	0.633	0.542	0.717
bind	0.442	0.350	0.525

B Background and formal setup

I use Stack Theory to talk about what an organism can represent and what it can learn. This section fixes notation and states the one prior used to score policies in Experiment 1. The experiments use the same objects via the Stack Theory suite code bundled with the driver.

B.1 Environments, programs, and vocabularies

Definition 1 (Environment, programs, and vocabularies)**.** *An* environment *is a nonempty set Φ of mutually exclusive states. A* (declarative) program *is any set of states $p \subseteq \Phi$. Let $\mathcal{P} := 2^{\Phi}$ denote the set of all programs. A* vocabulary *is any set of programs $\mathfrak{v} \subseteq \mathcal{P}$ that a system can implement. Unless specified otherwise, vocabularies are finite.*

Intuition. Think of Φ as the set of all possible objective realities. A state is one such reality. A program is a constraint on realities. A vocabulary is the system's stock of available constraints, set by its embodiment.

Finiteness in the experiments. The definitions above do not assume Φ is finite. In Experiments 1 to 3 I sample finite Φ and finite $\mathfrak{v}$ so that I can enumerate $L_{\mathfrak{v}}$ and count extensions. In the experiments, statements are represented as finite bitmasks, even though the formal definition of $L_{\mathfrak{v}}$ allows infinite subsets when the intersection is well-defined.

B.2 Statements, truth sets, and induced language

Definition 2 (Abstraction layer, statements, and truth sets)**.** *A vocabulary $\mathfrak{v}$ induces an embodied language*

$$L_{\mathfrak{v}} = \{\, \ell \subseteq \mathfrak{v} \mid \bigcap_{p \in \ell} p \neq \emptyset \,\}.$$

*Elements $\ell \in L_{\mathfrak{v}}$ are called **statements**. They behave like conjunctions of vocabulary programs. The **truth set** of ℓ is*

$$T(\ell) := \bigcap_{p \in \ell} p \subseteq \Phi,$$

*so ℓ is **true** at $\phi \in \Phi$ iff $\phi \in T(\ell)$, with the convention $T(\emptyset) := \Phi$.*

Intuition. A statement is a bundle of constraints that must all hold at once. Its truth set is the region of state space where that bundle is jointly satisfied.

B.3 Completions, extensions, equivalence, and weakness

Definition 3 (Completions, extensions, and equivalence)**.** *A **completion** of a statement $x \in L_{\mathfrak{v}}$ is any statement $y \in L_{\mathfrak{v}}$ such that $x \subseteq y$. The **extension** of x is the set of its completions*

$$E_x := \{y \in L_{\mathfrak{v}} \mid x \subseteq y\}.$$

For a set $X \subseteq L_{\mathfrak{v}}$, its extension is

$$E_X := \bigcup_{x \in X} E_x.$$

*Two statements x and y are **equivalent**, written $x \sim y$, iff $E_x = E_y$.*

Definition 4 (Weakness)**.** *The **weakness** of a statement $x \in L_{\mathfrak{v}}$ is*

$$w(x) := |E_x|.$$

Remark 1. *Weakness compares only cardinalities. In general, $w(x_1) \geq w(x_2)$ does not imply $E_{x_2} \subseteq E_{x_1}$. However, if $x_1 \subseteq x_2$ then $E_{x_2} \subseteq E_{x_1}$ and hence $w(x_1) \geq w(x_2)$.*

Intuition. Weak statements leave many completions possible. Strong statements rule out many completions. Weakness counts how many completions a statement still allows.

B.4 Tasks, correct policies, and the two proxies

Definition 5 (Task and correctness)**.** *Fix a finite vocabulary $\mathfrak{v}$ and its induced language $L_{\mathfrak{v}}$. A $\mathfrak{v}$ task is a pair $\alpha = \langle I_\alpha, O_\alpha \rangle$ where $I_\alpha, O_\alpha \subseteq L_{\mathfrak{v}}$ satisfy the consistency condition*

$$\forall i \in I_\alpha \; \exists o \in O_\alpha \text{ such that } i \subseteq o.$$

*I interpret I_α as a set of evidence statements and O_α as the set of outputs that were correct in those episodes. A policy is a statement $\pi \in L_{\mathfrak{v}}$. The policy is **correct** for α iff*

$$O_\alpha = E_{I_\alpha} \cap E_\pi.$$

Let Π_α denote the set of correct policies for α.

Intuition. The evidence fixes what must happen in the regions of $L_{\mathfrak{v}}$ that have been touched so far. The consistency condition matches the master Stack Theory task definition and rules out malformed input output pairs. A correct policy agrees with that evidence and does not add extra outputs in those regions. This definition also lets evidence include negative constraints, which is how Experiment 1 encodes negative examples.

Definition 6 (Weakness and description length proxies)**.** *For a statement $\pi \subseteq \mathfrak{v}$ define its description length proxy as the number of conjuncts*

$$d(\pi) := |\pi|.$$

Given a finite nonempty candidate set $C \subseteq L_{\mathfrak{v}}$ I define two selectors.

- $\mathrm{wmax}(C)$ *returns a policy with maximum $w(\pi)$ and breaks ties by minimum $d(\pi)$.*
- $\mathrm{simpmax}(C)$ *returns a policy with minimum $d(\pi)$ and breaks ties by maximum $w(\pi)$.*

Ties after this are broken deterministically by a fixed ordering of statements.

Intuition. Weakness is a proxy for how well a policy will generalize under unknown future constraints. Description length is a proxy for how cheap a policy is to store or to search for.

Lemma 1 (Uniform parent extension model)**.** *Fix a task $\alpha = \langle I_\alpha, O_\alpha \rangle$ in an induced language $L_{\mathfrak{v}}$. Let Π_α be the set of policies that are correct for α. Define the seen region*

$$E_{I_\alpha} := \bigcup_{\ell \in I_\alpha} E_\ell$$

and the unseen region

$$U_\alpha := L_{\mathfrak{v}} \setminus E_{I_\alpha}.$$

For a correct policy $\pi \in \Pi_\alpha$, define

$$B_\pi := E_\pi \cap U_\alpha.$$

Assume the world generates a parent by sampling a random extra constraint set $P \subseteq U_\alpha$ uniformly. A policy π remains correct for that parent if and only if $P \subseteq B_\pi$. Under this model, the generalisation probability is

$$\mathbb{P}(\pi \textit{ generalises}) = 2^{|B_\pi| - |U_\alpha|}.$$

For correct policies, maximising weakness $|E_\pi|$ is equivalent to maximising $|B_\pi|$ and therefore maximises the generalisation probability.

Proof. The parent adds extra constraints by picking $P \subseteq U_\alpha$. The policy survives exactly when the parent picks no constraint outside E_π, meaning $P \subseteq B_\pi$. There are $2^{|U_\alpha|}$ equally likely subsets of U_α and $2^{|B_\pi|}$ of them lie inside B_π. So the success probability is $2^{|B_\pi|}/2^{|U_\alpha|}$.

Now take two correct policies π_1, π_2. Because both are correct, $E_{\pi_i} \cap E_{I_\alpha} = O_\alpha$. So $|E_{\pi_i}| = |O_\alpha| + |B_{\pi_i}|$. The term $|O_\alpha|$ is the same for both policies. So maximising weakness $|E_\pi|$ is the same as maximising $|B_\pi|$. That also maximises the probability above. □

Proof intuition. The child task pins down what must happen in the seen region. Weakness is about what remains unpinned in the unseen region. Under a uniform parent model, each extra unseen constraint is just as likely as any other. So the best hedge is to keep the biggest safe region in the unseen part. This is the same counting argument used in [Bennett, 2023, 2025a].

C Causal identities

C.1 Interventions and observations

Fix an organism $\mathfrak{o}$. Let $\mathfrak{v}_{\mathfrak{o}}$ be its vocabulary and let $L_{\mathfrak{o}} := L_{\mathfrak{v}_{\mathfrak{o}}}$ be its induced language. Let $\mathrm{INT}_{\mathfrak{o}} \subseteq L_{\mathfrak{o}}$ be a set of statements describing self generated episodes. Let $\mathrm{OBS}_{\mathfrak{o}} \subseteq L_{\mathfrak{o}}$ be a set of statements describing non self episodes.

Definition 7 (Causal identity candidate)**.** *A statement $c \in L_{\mathfrak{o}}$ is a* causal identity candidate *for $\mathfrak{o}$ if*

- *c is entailed by every intervention episode*

$$c \subseteq I \quad \textit{for all } I \in \mathrm{INT}_{\mathfrak{o}}$$

- *c is not entailed by any observation episode*

$$c \not\subseteq O \quad \textit{for all } O \in \mathrm{OBS}_{\mathfrak{o}}$$

Let $\mathcal{C}_{\mathfrak{o}}$ denote the set of all candidates.

Intuition. A causal identity candidate is an invariant of the organism's own interventions that does not also hold for merely observed events. It is an internal handle for attribution. This mirrors efference and reafference distinctions in biology, and is functionally equivalent to a Pearlean *do* operator [Jékely et al., 2021, Pearl, 2009].

Definition 8 (w-maximised causal identity)**.** *Assume $\mathcal{C}_{\mathfrak{o}}$ is nonempty. Define the w-maximised causal identity of $\mathfrak{o}$ as*

$$c_{\mathfrak{o}}^{\mathfrak{o}} := \mathrm{wmax}(\mathcal{C}_{\mathfrak{o}}).$$

Intuition. Many invariants might separate interventions from observations. The weakness maximiser picks the one that makes the fewest additional commitments. It is the most robust attribution handle.

D The stack theoretic hierarchy of selves

D.1 Chain notation

In the aforementioned related work, causal identities are nested [Bennett et al., 2024, Bennett, 2025b]. I write $c_{\mathfrak{a}}^{\chi}$ for a causal identity represented inside organism $\mathfrak{a}$. The superscript χ is a finite chain of organisms. The intended reading is that longer chains encode deeper nested attributions. I only need the following special cases.

Definition 9 (First, second, and third order selves)**.** *Fix an organism $\mathfrak{a}$ and an audience organism $\mathfrak{b}$. Define*

$$\begin{aligned}
\mathfrak{a}_1 &:= c_{\mathfrak{a}}^{\mathfrak{a}} \\
\mathfrak{a}_2 &:= c_{\mathfrak{a}}^{\mathfrak{b}\mathfrak{a}} \\
\mathfrak{a}_3 &:= c_{\mathfrak{a}}^{\mathfrak{b}\mathfrak{a}\mathfrak{b}\mathfrak{a}}.
\end{aligned}$$

Intuition. $\mathfrak{a}_1$ is $\mathfrak{a}$'s own attribution handle for its interventions. $\mathfrak{a}_2$ is $\mathfrak{a}$'s model of how an audience attributes $\mathfrak{a}$. It captures the difference between acting and being seen acting. $\mathfrak{a}_3$ adds one more round of nesting. It is the minimal structure that supports deliberate self-binding in response to anticipated audience updates.

D.2 Decoder mismatch as a Bayesian failure mode

Theorem 1 (Decoder mismatch breaks passive attribution)**.** *Let an evaluator observe a trace y in some channel alphabet. Let its hypothesis class be $\mathcal{H} = \mathcal{H}_{\mathrm{mind}} \cup \{h_{\mathrm{noise}}\}$. Assume $\mathbb{P}(h_{\mathrm{noise}}) > 0$. Assume also $\mathbb{P}(y \mid h_{\mathrm{noise}}) > 0$. If there exists $\varepsilon \in (0,1)$ such that for every $h \in \mathcal{H}_{\mathrm{mind}}$*

$$\mathbb{P}(y \mid h) \leq \varepsilon\, \mathbb{P}(y \mid h_{\mathrm{noise}}),$$

then the posterior mass on mind hypotheses satisfies

$$\mathbb{P}(\mathcal{H}_{\mathrm{mind}} \mid y) \leq \frac{\varepsilon\, \mathbb{P}(\mathcal{H}_{\mathrm{mind}})}{\varepsilon\, \mathbb{P}(\mathcal{H}_{\mathrm{mind}}) + \mathbb{P}(h_{\mathrm{noise}})}.$$

In particular, if the evaluator's decoder family makes ε small, then passive attribution of mindedness becomes arbitrarily unlikely.

Intuition. If all mind-like explanations in your model class assign the trace far lower likelihood than a noise model, then Bayes rule forces the posterior away from mind. Decoder mismatch is one way to make that likelihood gap systematic [Bennett, 2022, Li and Vitányi, 2008].

Proof. Write

$$N := \sum_{h \in \mathcal{H}_{\text{mind}}} \mathbb{P}(h)\,\mathbb{P}(y \mid h) \quad \text{and} \quad A := \mathbb{P}(h_{\text{noise}})\,\mathbb{P}(y \mid h_{\text{noise}}).$$

Then

$$\mathbb{P}(\mathcal{H}_{\text{mind}} \mid y) = \frac{N}{N + A}.$$

By the assumption, for every $h \in \mathcal{H}_{\text{mind}}$ it holds that

$$\mathbb{P}(y \mid h) \leq \varepsilon\,\mathbb{P}(y \mid h_{\text{noise}}).$$

Summing with weights $\mathbb{P}(h)$ gives

$$N \leq \varepsilon\,\mathbb{P}(y \mid h_{\text{noise}})\,\mathbb{P}(\mathcal{H}_{\text{mind}}).$$

The function $x \mapsto x/(x + A)$ is increasing in x when $A > 0$. So the posterior is upper bounded by replacing N by its upper bound. This yields

$$\mathbb{P}(\mathcal{H}_{\text{mind}} \mid y) \leq \frac{\varepsilon\,\mathbb{P}(\mathcal{H}_{\text{mind}})\,\mathbb{P}(y \mid h_{\text{noise}})}{\varepsilon\,\mathbb{P}(\mathcal{H}_{\text{mind}})\,\mathbb{P}(y \mid h_{\text{noise}}) + \mathbb{P}(h_{\text{noise}})\,\mathbb{P}(y \mid h_{\text{noise}})}.$$

Cancel the common factor $\mathbb{P}(y \mid h_{\text{noise}})$. □

Proof intuition. The proof is a direct likelihood ratio bound. If mind models never fit the trace much better than noise, the posterior cannot concentrate on mind. A mismatched decoder family can enforce that condition even when the system is highly structured internally.

D.3 First-order self is unavoidable for causal control

Theorem 2 (Intervention tagging is unavoidable)**.** *Let an organism $\mathfrak{o}$ update an internal state s after each episode and select actions as a function of that state. Assume the organism can improve expected utility by conditioning its behaviour on whether an episode was generated by its own intervention or was merely observed, while holding fixed the episode content as represented in $L_{\mathfrak{o}}$. Then there exist two reachable internal states that differ only in this intervention versus observation status. Equivalently, the organism's state must encode an intervention tag.*

Intuition. Causal control requires distinguishing do from see. Without that distinction, the agent cannot represent the effect of its own actions [Pearl, 2009].

Proof. Assume for contradiction that the organism's post-episode state is a function only of the episode content statement in $L_{\mathfrak{o}}$. Then any intervention episode and observation episode that map to the same content statement lead to the same internal state. A policy that depends only on internal state then cannot condition its behaviour on the intervention versus observation status in those cases.

By assumption, such conditioning can improve expected utility. So there exist two episodes with the same represented content but different causal status for which the optimal continuation differs. That requires the internal state to differ across the two cases. This contradicts the assumption that the post-episode state depends only on content. Therefore the internal state must encode an intervention tag. □

Proof intuition. If an agent can act differently depending on whether something was done by it or merely observed, then the information that distinguishes those cases must exist in its state. This is the minimal content of an efference copy [Jékely et al., 2021].

Theorem 3 (First-order self as the weakest separator)**.** *Assume $\mathcal{C}_{\mathfrak{o}}$ is nonempty. Let $c^{\mathfrak{o}}_{\mathfrak{o}}$ be a w-maximised causal identity as in Definition 8. In the conjunctive language $L_{\mathfrak{o}}$, there is no candidate $c' \in \mathcal{C}_{\mathfrak{o}}$ with $c' \subset c^{\mathfrak{o}}_{\mathfrak{o}}$.*

Proof. Let $c^* := c^{\mathfrak{o}}_{\mathfrak{o}} = \mathrm{wmax}(\mathcal{C}_{\mathfrak{o}})$. Suppose for contradiction that there exists a candidate $c' \in \mathcal{C}_{\mathfrak{o}}$ with $c' \subset c^*$. Because $c' \subset c^*$, every completion of c^* is also a completion of c'. So $E_{c^*} \subseteq E_{c'}$ and therefore $w(c') \geq w(c^*)$. But c^* maximises w over $\mathcal{C}_{\mathfrak{o}}$, so also $w(c') \leq w(c^*)$. Hence $w(c') = w(c^*)$.

Because $c' \subset c^*$, it follows that $d(c') = |c'| < |c^*| = d(c^*)$. Definition 6 says that wmax breaks ties in favour of smaller d. So $\mathrm{wmax}(\mathcal{C}_{\mathfrak{o}})$ could not return c^*. This is a contradiction. □

Proof intuition. If there were a strictly smaller separator, it would make fewer commitments. It would be compatible with at least as many completions. That would match or exceed the weakness of the supposed w-maximiser. But then the chosen identity could not be minimal among w-maximisers.

D.4 Second-order self is necessary for Gricean self-report

Definition 10 (Gricean self-report)**.** *A* Gricean self-report *by organism $\mathfrak{a}$ is an utterance policy that is chosen in part to shape what a listener $\mathfrak{b}$ will infer about $\mathfrak{a}$. The report succeeds when $\mathfrak{b}$'s inferred causal identity of $\mathfrak{a}$ matches the aspect of $\mathfrak{a}$ that $\mathfrak{a}$ intended to convey.*

Intuition. A self-report is not just a statement about the world. It is a statement chosen to cause a listener to update a model of the speaker. This is the standard Gricean move [Grice, 1957, 1969].

Theorem 4 (Second-order self is necessary for Gricean report)**.** *Consider a communication game where a speaker $\mathfrak{a}$ must transmit a private bit $m \in \{0, 1\}$ to a listener $\mathfrak{b}$. The listener type $t \in \{0, 1\}$ is drawn uniformly at random, independent of m, and is hidden from $\mathfrak{a}$. If $t = 0$ then $\mathfrak{b}$ decodes the signal literally. If $t = 1$ then $\mathfrak{b}$ flips the decoded bit. Assume $\mathfrak{a}$ can send a probe signal and observe one listener response before sending the final signal.*

Any speaker policy that cannot condition on the listener response achieves expected success at most $\frac{1}{2}$. There exists a policy that conditions on the response and achieves success probability 1. Such conditioning requires a second-order self state $\mathfrak{a}_2$.

Intuition. With two possible audience decoders, the speaker has to infer which decoder is active. A single observed response is enough to do that. If the speaker cannot represent that inference, it cannot do better than guessing.

Proof. If the speaker policy cannot condition on the observed response, then the final signal is a deterministic or randomized function of m alone. Conditioned on m, half the time the listener type is $t = 0$ and half the time it is $t = 1$. For any fixed final signal distribution, the probability of correct decoding under $t = 0$ is exactly the probability of incorrect decoding under $t = 1$. Averaging over the two types yields success probability $\frac{1}{2}$.

Now consider the policy that sends probe $p = 0$. If the observed listener response equals 0 infer $t = 0$. Otherwise infer $t = 1$. Then send final signal $s = m$ when $t = 0$ and $s = 1 - m$ when $t = 1$. This forces the listener to decode m correctly in either case.

The policy depends on the inferred listener type. Representing that dependence is exactly the role of an internal state that models how the listener will interpret the speaker's signals. This is the minimal content of a second-order self in Stack Theory. □

Proof intuition. Without a model of the listener, the speaker cannot tailor the message. With a model of the listener, a short interaction can identify the decoder and enable reliable report. In this toy game, the model is simply a posterior over a finite decoder set. This is audience dependence in its simplest formal form.

D.5 Third-order self is necessary for trust equilibria

Theorem 5 (Third-order self is necessary for trust equilibria)**.** *Consider the one-shot trust game. Player* $\mathfrak{a}$ *moves first and chooses whether to bind itself by publicly committing to a restricted action set. Player* $\mathfrak{b}$ *then chooses whether to trust. If* $\mathfrak{b}$ *trusts, then* $\mathfrak{a}$ *chooses between* honor *and* exploit *when exploit is available. Honor gives both players payoff* 1*. Exploit gives* $\mathfrak{a}$ *payoff* $1 + g$ *and gives* $\mathfrak{b}$ *payoff* -1 *for some* $g > 0$*. Binding removes exploit and imposes a cost* c *on* $\mathfrak{a}$*.*

If binding is not available, the unique subgame perfect equilibrium has no trust. If binding is available and $c < 1$*, there exists a subgame perfect equilibrium in which* $\mathfrak{a}$ *binds,* $\mathfrak{b}$ *trusts, and the outcome payoff is* $\langle 1 - c, 1\rangle$*. Executing the binding move requires a third-order self state* $\mathfrak{a}_3$*.*

Intuition. Without commitment, exploitation is optimal for the trustee. The trustor anticipates that and refuses to trust. If the trustee can bind itself, trust can become rational. The binding move is a policy over the future self. It depends on anticipating how the audience will update. That is third-order self structure [Schelling, 1960, Elster, 1979].

Proof. First assume binding is not available. If $\mathfrak{b}$ trusts, then $\mathfrak{a}$ chooses between honor with payoff 1 and exploit with payoff $1 + g$. Since $g > 0$, exploit strictly dominates honor. Therefore in any subgame perfect equilibrium, $\mathfrak{a}$ exploits after trust. Anticipating this, $\mathfrak{b}$ obtains payoff -1 from trusting and 0 from not trusting. Therefore $\mathfrak{b}$ does not trust. This yields the unique subgame perfect equilibrium outcome with payoff $\langle 0, 0\rangle$.

Now assume binding is available and consider the following strategy profile. Player $\mathfrak{a}$ binds. Player $\mathfrak{b}$ trusts if and only if $\mathfrak{a}$ bound. If $\mathfrak{b}$ trusts and exploit is unavailable, then $\mathfrak{a}$ honors.

Given binding, $\mathfrak{a}$ cannot exploit. Honor yields payoff $1 - c$. If $c < 1$, then $1 - c > 0$. Not binding leads to no trust and payoff 0. So $\mathfrak{a}$ prefers to bind. Given binding, $\mathfrak{b}$ expects honor and obtains payoff 1. So $\mathfrak{b}$ prefers to trust. Off the binding path, if $\mathfrak{a}$ does not bind then exploit is available. By the first part, $\mathfrak{b}$ rationally refuses to trust. Therefore the strategy profile is subgame perfect.

Finally, implementing the binding move requires $\mathfrak{a}$ to choose a constraint on its own future action set. That is a policy about the policy that a future self will execute. It is naturally represented as an additional nesting level beyond $\mathfrak{a}_2$. □

Proof intuition. Binding is valuable because it changes what the other player expects the agent will do. To decide to bind, the agent must model that expectation update. To then follow through, the agent must have removed the tempting action from its future options.

D.6 Convergence under w-maximisation

Theorem 6 (Self convergence under w-maximisation)**.** *Fix an organism* $\mathfrak{o}$ *and a viability task* α*. Assume that among all correct policies for* α*, every correct policy implies a particular self statement* s*. Assume also that* s *itself is a correct policy for* α*. Then any w-maximising learning rule that searches over correct policies can converge to* s*.*

Intuition. If a self statement is necessary for viability, then every viable policy contains it. If the self statement alone is already sufficient, it is the weakest viable policy. So a w-maximiser will prefer it.

Proof. Let Π be the set of policies that are correct for task α. By assumption, $s \subseteq \pi$ for every $\pi \in \Pi$. Then for every $\pi \in \Pi$, $E_\pi \subseteq E_s$. This implies $w(\pi) \leq w(s)$. Since $s \in \Pi$, s is a w-maximiser. A learning rule that searches Π and selects a w-maximiser can therefore converge to s. □

Proof intuition. If every viable policy contains the same self core, the only remaining choice is how much extra structure to add. Weakness maximisation says do not add any extra structure unless it is necessary.

E Methods and experiments

All experiments are randomised Monte Carlo benchmarks. The goal is to average over external factors. This makes it harder to smuggle in fixed structure.

E.1 Experiment 1 Structured randomised abstraction layers

World generator. Each world samples a finite environment Φ with $n \in [32, 64]$ mutually exclusive states and a vocabulary size $|\mathfrak{v}| \in [9, 13]$. I represent each environment state by a *state encoding* which is a maximal consistent statement mask. The backbone features are fixed indices in every world. They are j, k, z, u_j, and u_k. I then build the state encodings from a small set of templates. Training states always contain z, j, and k. Core states contain j and k but not z. Negative evidence states come in two kinds. The u_j kind contains j and u_j but not k or z. The u_k kind contains k and u_k but not j or z. Background states contain only auxiliary features. This construction makes z observationally indistinguishable from $j \wedge k$ on the child task, but separable by parent extensions. Given the list of state encodings, I define each program p_i as the set of state indices whose encoding contains feature i. This produces a vocabulary $\mathfrak{v}$ over Φ. The induced language $L_{\mathfrak{v}}$ is then the set of all satisfiable statements, meaning all statement masks with nonempty truth set.

Task sampler. The hidden target is $\pi_* = j \wedge k$. I sample one training state whose encoding contains z, j, and k, plus at least one auxiliary feature. Let m_{train} be that state encoding. The task output set is $O = \{m_{\text{train}}\}$. The input set contains three statements. It contains m_{train} as the positive training input. It contains the singleton statements u_j and u_k as negative evidence inputs. Given inputs I and output O, the suite task object uses the correctness equation $E_I \cap E_\pi = O$. Here E_I is the union of the completion sets of the individual input statements.

Selectors. On each task I enumerate the correct policies. wmax selects a correct policy with maximal weakness $|E_\pi|$ and breaks ties by simplicity. $\mathrm{simpmax}$ selects a correct policy with minimal description length and breaks ties by weakness. A random baseline selects uniformly from the set of correct policies.

Parent model scoring. Let $U = L_{\mathfrak{v}} \setminus E_I$ be the unseen region. A parent task adds an unseen subset $S \subseteq U$ to the child output set, giving $O_\omega = O \cup S$. I evaluate selectors by their log probability of remaining correct under this parent extension. Under the uniform parent model, S is sampled uniformly from the power set of U. Under the nonuniform model, each $u \in U$ is included independently with probability q_u. I draw q_u i.i.d. from a Beta(a, b) distribution. The reported default uses $(a, b) = (0.7, 2.0)$, and Table 1 sweeps across a small set of (a, b) pairs.

Aggregation. I run 30 worlds with 20 accepted tasks per world. For the nonuniform parent, I draw 8 independent q vectors per task. I report world means and 95% bootstrap confidence intervals across worlds.

E.2 Experiment 2 Decoder mismatch with probing

World generator. Each world samples an alphabet size $m \in [4, 8]$ and a candidate decoder set of $k \in [4, 12]$ random permutations of the alphabet. A noise rate $\varepsilon \in [0, 0.10]$ is sampled uniformly. The training decoder is one element of the candidate set and remains fixed. In the stationary setting, the test decoder is another element fixed across episodes. In the nonstationary setting, the test decoder is resampled each episode from the same candidate set.

Game and strategies. The speaker must transmit a random target symbol. It may send p probe symbols and observe the decoded outputs. It then sends a final symbol. Passive assumes the training decoder. Random probing chooses probes uniformly. Information seeking probing chooses probes that maximise expected output entropy under the current posterior. Oracle knows the decoder.

Run configuration. This draft reports 60 worlds. Each world evaluates 150 independent episodes for each probe count $p \in \{0, 1, 2, 3\}$.

E.3 Experiment 3 Trust with binding

World generator. Each world samples payoffs for a finite horizon trust game and a one time binding cost. Payoffs satisfy the usual ordering. Cooperation is Pareto preferred for the pair. Exploitation benefits the agent and harms the observer. No trust yields an intermediate baseline.

Equilibrium model. Without binding, backward induction implies exploitation. So a rational observer refuses to trust. With binding, the system binds when $HR_s - c > HP_s$, equivalently when $H(R_s - P_s) > c$. The observer then trusts because exploitation is infeasible.

Run configuration. This draft reports 120 worlds and horizons $H \in \{1, 2, 3, 4, 5\}$.

E.4 Reproducibility checklist

1. Run `python NSNC_Experiments.py --outdir run1 --which all --seed 11`.
2. The script writes figures to `outdir/figures/`, tables to `outdir/tables/`, and raw outputs to `outdir/results/`.
3. The file `outdir/results/all_experiments_summary.json` includes the SHA256 hash of the script and the full run configuration.
4. The Stack Theory Suite code used by the driver is bundled under `stacktheory_suite/`. This draft uses Suite release 0.7.2.

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